Massimo Perfetti

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EDUCATION

Massachusetts Institute of Technology (MIT) *Cambridge, MA | Apr. 2026 - Present*
Master's Thesis in Mechanical Engineering and Computer Science
- **Research Topics:** State-Space Models, Deep Learning, Reduced-Order Modeling, Fluid Dynamics, UQ.
- **Activities:** Hack-Nation 6th Global AI Hackaton winner (5500+ applicants).

Delft University of Technology (TU Delft) *Delft, Netherlands | Sept. 2024 - Dec. 2026*
Master's in Mechanical Engineering and Computer Science
- **Honours:** HPM - TU Delft Honours Program *(reserved to 100 MSc students every year).*
- **Relevant Coursework:** Deep Learning, Advanced Fluid Dynamics, Computational Fluid Dynamics, Reinforcement Learning, Generative Modeling, Machine Learning for Graph Data.

EPFL & Polytechnic University of Turin *Turin, Italy & Lausanne, Switzerland | Sep. 2021 – Jul. 2024*
Bachelor's in Mechanical Engineering
- **Honours:** Percorso Intraprendenti - PoliTO Honours Program *(reserved to top 240 students)*

WORK EXPERIENCE

Visiting Researcher | Van Rees Lab, MIT *Cambridge, Ma | Apr. 2026 - Present*
Research Overview: M.Sc. Thesis at the intersection of CFD and machine learning, developing sequence-based ML architectures trained on high-fidelity flow maps to predict and exploit wind conditions for Urban Air Mobility (UAM) navigation under the supervision of Professor W. van Rees.

CTO & Co-Founder | Dual Energy *Sep. 2025 – Present*
About: Dual Energy is a B2B startup building the intelligence layer of artificial intelligence for the agrivoltaics.
Role: Responsible for the full technical stack, driving product roadmap and AI research and design.
Official website: https://www.dualenergy.ch

R&D Intern | Novelis Switzerland SA *Sierre, Switzerland | Sep. 2025 – Feb. 2026*
Project Overview: Developed CAD + CFD models for transient multiphase fluid flow in melt treatment systems, implementing custom solvers with OpenFOAM (C++), and validating against CFD software.

AI Research Intern | AIdro Lab TU Delft *Delft, Netherlands | Mar. 2025 – Jun. 2025*
Research Aim: Fine-tuning of foundational AI models (SWE-GNNs) for spatio-temporal flood modelling under the supervision of Professor R. Taormina, heavily relying on NVIDIA PhysicsNeMo framework.

PROJECTS

Unidiary.ai *Oct. 2026 – Feb. 2026*
Project Story: Co-founded and built UniDiary.ai, an AI study-planning web app for university students, owning the full product and technical stack; ran a one-month pilot at EPFL reaching a couple hundred students. (https://www.unidiary.ai)

Wind Power Forecasting with GNNs *Apr. 2025 – Jun. 2025*
Project Focus: Scalable Graph Convolutional Network (GCN) architectures for short-term, turbine-level wind power forecasting by capturing spatio-temporal dependencies, implementing a flexible pipeline on the SDWPF dataset (https://github.com/Chefcurry4/GML).

SKILLS, ACTIVITIES & INTERESTS

Languages: English (C2), Italian (native), French (B2).
Skills: Python, Pytorch, Deep Learning, CFD & HPC, CAD, Docker, PostgreSQL, C++, Git, LLM tooling.
Interests: NBA, Chess, Machine Learning, Logic games, CFD, Full Stack Development, Start Up, AI agents.